Wireshark SOC Lab Project – Network Traffic Analysis

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1. Project Overview

This project demonstrates practical experience in network traffic analysis using Wireshark. The objective was to capture, inspect, and analyse DNS and HTTP traffic within a virtualised Ubuntu environment, and determine whether the observed activity was legitimate or suspicious.

2. Lab Environment

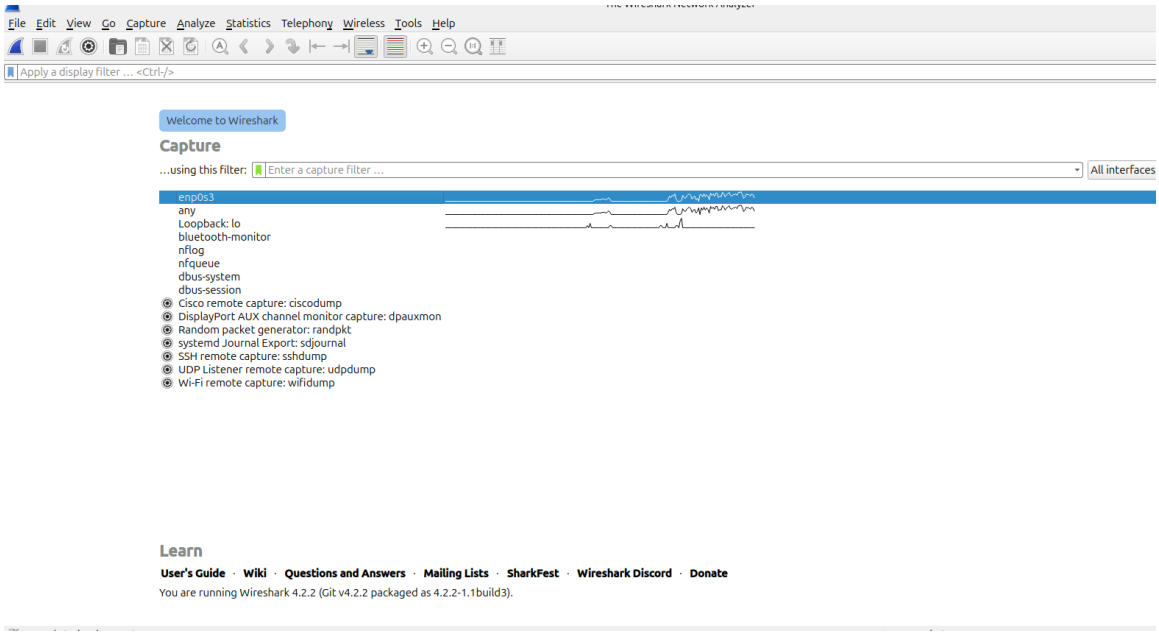
- Ubuntu 24.04 (Virtual Machine - VirtualBox)
- Wireshark
- Firefox Browser
- Network Interface: enp0s3

3. Methodology

Wireshark was used to capture live network traffic. DNS and HTTP filters were applied to isolate relevant packets. Web activity was generated through Firefox to simulate normal user behaviour. Selected packets were analysed using the TCP stream feature to inspect full request-response communication.

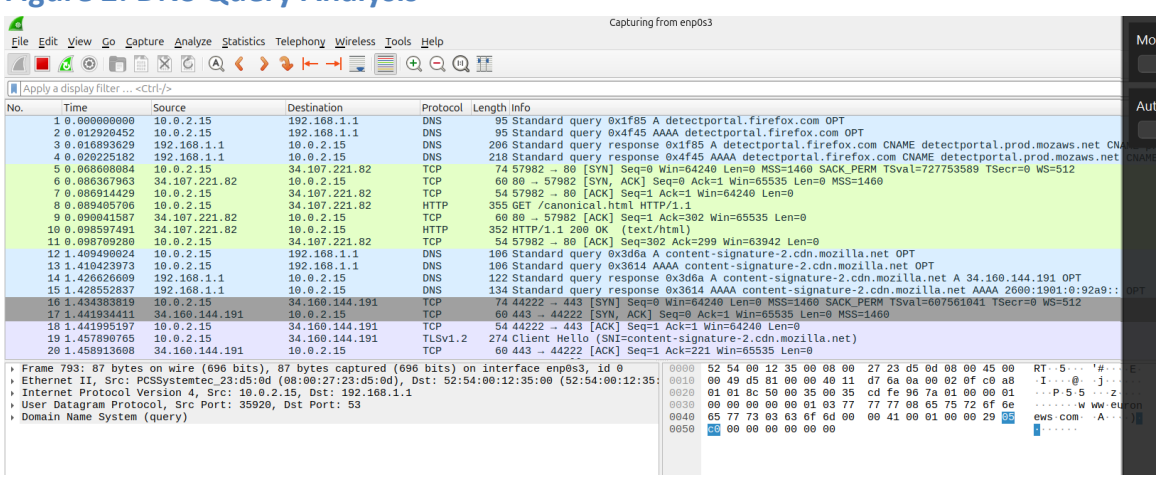
4. Evidence / Screenshots

Figure 1: Initial Packet Capture



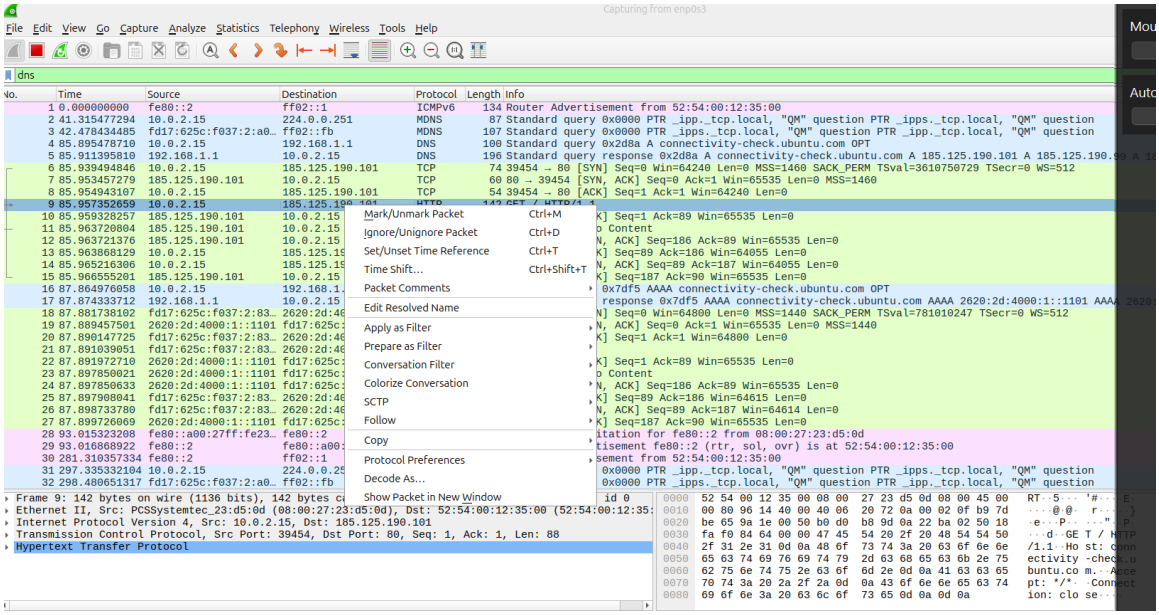
This screenshot shows live packet capture on interface enp0s3 with multiple protocols including DNS, TCP, and ICMP traffic.

Figure 2: DNS Query Analysis



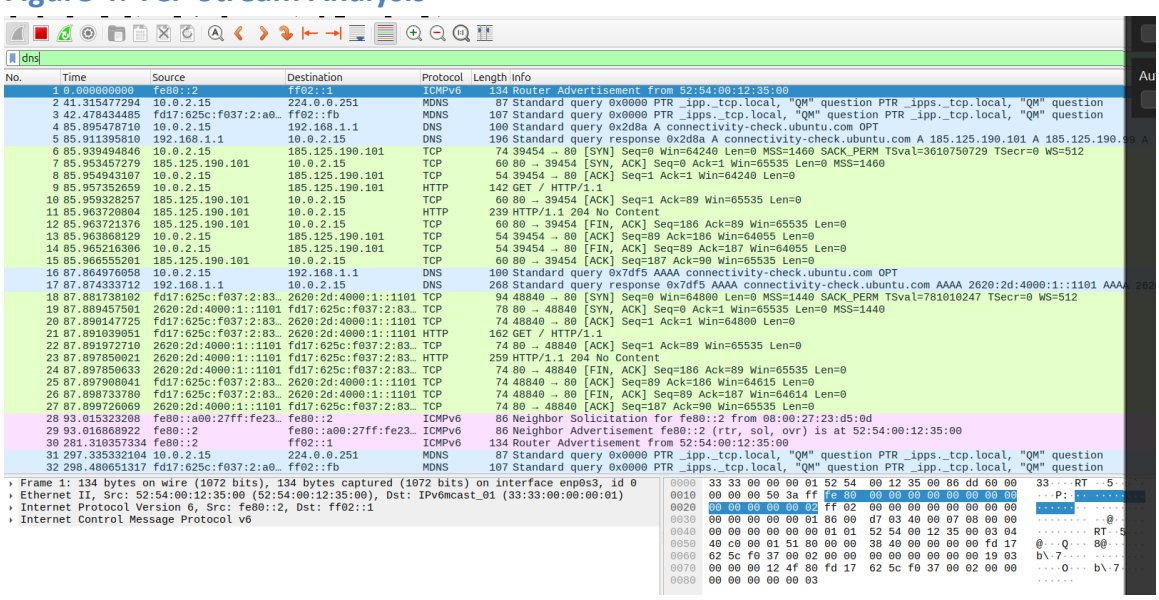
This shows DNS queries to detectportal.firefox.com, which is used by Firefox to verify internet connectivity.

Figure 3: HTTP Traffic



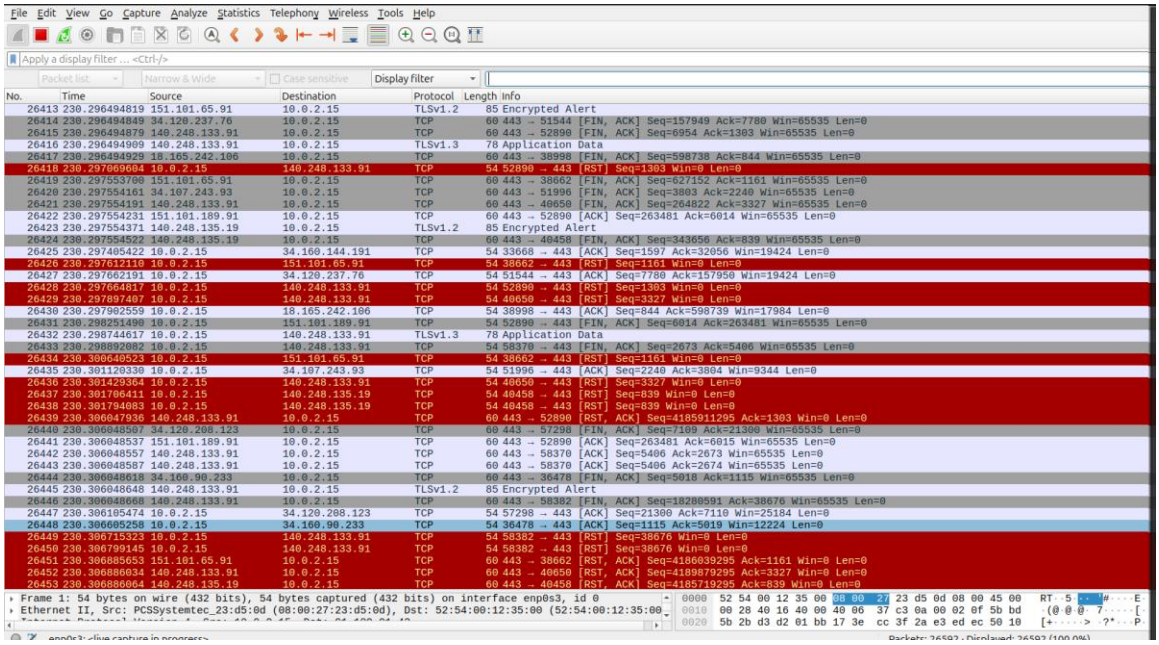
This screenshot highlights HTTP GET requests and responses, showing communication between the local machine and external servers.

Figure 4: TCP Stream Analysis



This demonstrates packet inspection and deeper analysis using Wireshark tools to understand full communication flow.

Figure 5: Mixed Traffic Analysis



No.	Time	Source	Destination	Protocol	Length	Info
26413	230.296494819	151.101.65.91	10.0.2.15	TLSv1.2	85	Encrypted Alert
26414	230.296494849	34.120.237.76	10.0.2.15	TCP	60	443 → 51544 [FIN, ACK] Seq=157949 Ack=7780 Win=65535 Len=0
26415	230.296494879	140.248.133.91	10.0.2.15	TCP	60	443 → 52890 [FIN, ACK] Seq=6954 Ack=1303 Win=65535 Len=0
26416	230.296494909	140.248.133.91	10.0.2.15	TLSv1.3	78	Application Data
26417	230.296494929	18.165.242.106	10.0.2.15	TCP	60	443 → 38998 [FIN, ACK] Seq=598738 Ack=844 Win=65535 Len=0
26418	230.297009604	10.0.2.15	140.248.133.91	TCP	54	52890 → 443 [RST] Seq=1303 Win=0 Len=0
26419	230.297553700	151.101.65.91	10.0.2.15	TCP	60	443 → 38662 [FIN, ACK] Seq=627152 Ack=1161 Win=65535 Len=0
26420	230.297554161	34.107.243.93	10.0.2.15	TCP	60	443 → 51996 [FIN, ACK] Seq=3803 Ack=2240 Win=65535 Len=0
26421	230.297554191	140.248.133.91	10.0.2.15	TCP	60	443 → 40650 [FIN, ACK] Seq=264822 Ack=3327 Win=65535 Len=0
26422	230.297554231	151.101.189.91	10.0.2.15	TCP	60	443 → 52890 [ACK] Seq=263481 Ack=6014 Win=65535 Len=0
26423	230.297554371	140.248.135.19	10.0.2.15	TLSv1.2	85	Encrypted Alert
26424	230.297554522	140.248.135.19	10.0.2.15	TCP	60	443 → 40458 [FIN, ACK] Seq=343656 Ack=839 Win=65535 Len=0
26425	230.297405422	10.0.2.15	34.160.144.191	TCP	54	33668 → 443 [ACK] Seq=1597 Ack=32056 Win=19424 Len=0
26426	230.297612110	10.0.2.15	151.101.65.91	TCP	54	38662 → 443 [RST] Seq=1161 Win=0 Len=0
26427	230.297662191	10.0.2.15	34.120.237.76	TCP	54	51544 → 443 [ACK] Seq=7780 Ack=157950 Win=19424 Len=0
26428	230.297664817	10.0.2.15	140.248.133.91	TCP	54	52890 → 443 [RST] Seq=1303 Win=0 Len=0
26429	230.297897407	10.0.2.15	140.248.133.91	TCP	54	40650 → 443 [RST] Seq=3327 Win=0 Len=0
26430	230.297902559	10.0.2.15	18.165.242.106	TCP	54	38998 → 443 [ACK] Seq=844 Ack=598739 Win=17984 Len=0
26431	230.298251499	10.0.2.15	151.101.189.91	TCP	54	52890 → 443 [FIN, ACK] Seq=6014 Ack=263481 Win=65535 Len=0
26432	230.298744617	10.0.2.15	140.248.133.91	TLSv1.3	78	Application Data
26433	230.298892082	10.0.2.15	140.248.133.91	TCP	54	58370 → 443 [FIN, ACK] Seq=2673 Ack=5406 Win=65535 Len=0
26434	230.300640523	10.0.2.15	151.101.65.91	TCP	54	38662 → 443 [RST] Seq=1161 Win=0 Len=0
26435	230.301120330	10.0.2.15	34.107.243.93	TCP	54	51996 → 443 [ACK] Seq=2240 Ack=3804 Win=9344 Len=0
26436	230.301429364	10.0.2.15	140.248.133.91	TCP	54	40650 → 443 [RST] Seq=3327 Win=0 Len=0
26437	230.301706411	10.0.2.15	140.248.135.19	TCP	54	40458 → 443 [RST] Seq=839 Win=0 Len=0
26438	230.301794083	10.0.2.15	140.248.135.19	TCP	54	40458 → 443 [RST] Seq=839 Win=0 Len=0
26439	230.306047936	140.248.133.91	10.0.2.15	TCP	60	443 → 52890 [RST, ACK] Seq=4185911295 Ack=1303 Win=0 Len=0
26440	230.306048507	34.120.208.123	10.0.2.15	TCP	60	443 → 57298 [FIN, ACK] Seq=7109 Ack=21300 Win=65535 Len=0
26441	230.306048537	151.101.189.91	10.0.2.15	TCP	60	443 → 52890 [ACK] Seq=263481 Ack=6015 Win=65535 Len=0
26442	230.306048557	140.248.133.91	10.0.2.15	TCP	60	443 → 58370 [ACK] Seq=5406 Ack=2673 Win=65535 Len=0
26443	230.306048587	140.248.133.91	10.0.2.15	TCP	60	443 → 58370 [ACK] Seq=5406 Ack=2674 Win=65535 Len=0
26444	230.306048018	34.160.90.233	10.0.2.15	TCP	60	443 → 36478 [FIN, ACK] Seq=5018 Ack=1115 Win=65535 Len=0
26445	230.306048648	140.248.133.91	10.0.2.15	TLSv1.2	85	Encrypted Alert
26446	230.306048668	140.248.133.91	10.0.2.15	TCP	60	443 → 58382 [FIN, ACK] Seq=18280591 Ack=38676 Win=65535 Len=0
26447	230.306105474	10.0.2.15	34.120.208.123	TCP	54	57298 → 443 [ACK] Seq=21300 Ack=7110 Win=25184 Len=0
26448	230.306605258	10.0.2.15	34.160.90.233	TCP	54	36478 → 443 [ACK] Seq=1115 Ack=5019 Win=12224 Len=0
26449	230.306715323	10.0.2.15	140.248.133.91	TCP	54	58382 → 443 [RST] Seq=38676 Win=0 Len=0
26450	230.306799145	10.0.2.15	140.248.133.91	TCP	54	58382 → 443 [RST] Seq=38676 Win=0 Len=0
26451	230.306805653	151.101.65.91	10.0.2.15	TCP	60	443 → 38662 [RST, ACK] Seq=4186639295 Ack=1161 Win=0 Len=0
26452	230.306886034	140.248.133.91	10.0.2.15	TCP	60	443 → 40650 [RST, ACK] Seq=4189879295 Ack=3327 Win=0 Len=0
26453	230.306886064	140.248.135.19	10.0.2.15	TCP	60	443 → 40458 [RST, ACK] Seq=4185719295 Ack=839 Win=0 Len=0

This screenshot shows multiple TCP reset packets and encrypted traffic, indicating normal secure communications and session terminations.

5. Key Findings

- DNS queries to detectportal.firefox.com were observed.
- HTTP GET requests identified
- Legitimate HTTP 200 OK responses
- Traffic confirmed as normal browser behaviour

6. Conclusion

The analysed traffic was legitimate and part of normal system operations. This project demonstrates core SOC analyst skills in traffic monitoring and analysis.